

BCSE204L Design and Analysis of Algorithms

Module-2

Topic-7: Branch and Bound

Branch and Bound

- Another algorithm design strategy
- Similar to backtracking because of usage of state space tree
- But branch and bound is for optimization problems only
- That too for minimization problems
- But maximization and minimization problems are interconvertible

Branch and Bound basics

- Unlike backtracking (which uses depth first search) branch and bound uses breadth first search to expand the nodes
- Comes in three varieties
- FIFO branch and bound (uses queue)
- LIFO branch and bound (uses stack)
- Least cost branch and bound

Core Idea

- Branching: Dividing the solution space
- Decision to be taken: To include a node or not
- Bounding: helps to prune (reduce) efficiently the tree, by killing nodes that do not satisfy bounding
- Leads to a faster exploration of the solution space
- Can be applied when greedy/DP fails

LIFO, FIFO B&B

- For the set $\{A, B, C, D\}$
- Demos: on board

Least Cost Branch and Bound

- In LIFO and FIFO B&B, the rule is rigid and blind
- The next node is expanded without giving any preferences to a node that has a very good chance of getting into an answer quickly

Fixed Solution set and Variable Solution set

- In variable solution, we will represent only the nodes that are selected. For eg: $\{A, B\}$, $\{A, C\}$ etc.
- In variable solution, use bit vectors. For eg: $\{A, B\} \rightarrow \{1, 1, 0, 0\}$, $\{A, C\} \rightarrow \{1, 0, 1, 0\}$ etc.

Thank you..